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AES Wastewater System in Grand Designs NZ



Grand Designs NZ will feature a house near Motueka, Nelson with a raised bed AES (Advanced Enviro-septic) wastewater system installation. The new series began this week on TV3 on Tuesdays, 7.30pm, with the Motueka house screening mid-October. The property is adjacent to an inlet where tidal water rises within 5m of the northern boundary on spring tides. Combined with a high water table this requires the AES pipes to be installed in a raised bed to provide 0.6m vertical separation to the water table. Pipework suspended under the building subfloor connects to the septic tank which is also partially raised allowing effluent to gravity flow to the AES bed.

The design is for a four bedroom house treating 1400 l/day. The photo above shows the AES bed prior to covering with topsoil and planting with coastal shrubs. The job was designed by Gary Stevens and installed by Fusion Plumbing. The homeowners are looking forward to the ongoing low-cost performance of their AES system.

When high groundwater is a design factor, a passive AES septic system can often be achieved by mounding up the ground around the septic tank and bed, and suspending drainage pipework to conserve elevation.

The photo below shows a similar AES raised bed two years after installation, at a property with winter groundwater at 450mm in winter. In this case the wastewater is pumped to the AES bed from the septic tank, which incorporates a pump well.



Latest Test Results for AES system in OSET-NTP Trial 12

ET installed an AES system at the Rotorua Onsite Effluent Treatment, National Treatment Programme (OSET-NTP) testing facility in September 2016. That system has taken part in trial 12 which ran from October 2016 – July 2017. Published reports are due early 2018. The test results have mirrored the advanced secondary results achieved in North America and Europe. The interim results from the benchmarking period are shown in the table below. We are retaining the system for Trial 13, where we will recirculate the treated effluent back through the anaerobic septic tank to further reduce total nitrogen (TN). Test results for Trial 12 can be viewed at our website et.nz.

- A+ rating for BOD5
- A+ for TSS
- D rating for Total Nitrogen
- A rating for Ammoniacal nitrogen
- B rating for Total Phosphorus
- B rating for Faecal Coliforms

Test	Median	Std Deviation
BOD (g/m ³)	2	1.45
TSS (g/m ³)	3	3.11
TN (g/m ³)	36.98	5.0
NH ₄ ⁻ (g/m ³)	1.67	0.5
TP (g/m ³)	2.59	0.4
FC (cfu/100mL)	5,730	6,624.3

Table 1. Interim results for the benchmarking period of the AES-38 system in Trial 12

A+ treatment quality for BOD and

TSS was also achieved after five days of the system operating at 200% of design loading, and an A+ rating for energy use as the only power required was as a consequence of having to pump to the effluent sampling chamber at the OSET-NTP facility.

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